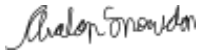


Semester (Term, Year)	Fall, 2025
Course Code	AER507
Course Section	Section 1
Course Title	Materials and Manufacturing
Course Instructor	Prof. Jeff Xi
Submission	
Submission No.	1
Submission Due Date	December 1st, 2025
Title	AER 507 Project Report
Submission Date	December 1st, 2025

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Abstract

A glider was to be designed with the goal of it to coast through the air for the longest time upon being hand thrown. Several constraints were allocated prior to the design process; some of which include the glider dimensions to be under 30 square inches along with additional limitations on the size of the fuselage, tail, in addition to the materials used for them. The team was tasked with applying aerodynamic principles and other knowledge learnt throughout various courses in aerospace engineering to maximize the payload to weight ratio as well as the gliding time of the aircraft. Upon understanding and getting a lucid understanding of the objective of the task, an objective tree as well as a pairwise comparison chart was created to visualize in addition to prioritize what needed to be done. Carbon fiber was chosen to be the material to be used for the wing and the internal structure was made of foam. Another design selection was that a skeletal structure was decided for the tail by the team to reduce weight and have sufficient room to have counter balances if needed. The manufacturing process as well as the design process were analyzed heavily in order to match with the prioritized objectives. Afterwards rough designs were made which were later finalized in the SolidWorks software. The flight was successful as the glider had steady and stable flights covering a good distance.

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1.0 Introduction

The goal of this project is to design and manufacture a hand thrown glider using appropriate materials and manufacturing processes. The design of the glider will be selected such that its flight distance to weight ratio is maximized, involving considerations such as its weight, ductility, and aerodynamics. These considerations will be evaluated based on the geometry of the major glider components, the materials selected, and the manufacturability of said materials. The design will be limited to the fuselage and tail, as well as the positioning of the wings, considering constraints such as the use of a composite wing, a maximum combined fuselage and wing weight of 175g, and various geometric constraints. Because a large focus of the design lies in the material and manufacturing processes selected, this report will provide insight on how to create an aerodynamic design practically.

2.0 Glider Design

The structural design of the glider is organized around three main components: the composite main wing, a lightweight display board fuselage and a wooden tail assembly. The wing planform is fixed by the project specifications, so the primary structural decisions focus on how the fuselage supports this wing and positions the tail at an appropriate moment arm for stability. The fuselage is sized to satisfy the overall length and width limits while providing a stiff backbone to carry bending loads from the wing and tail. Finally, the horizontal and vertical stabilizers are designed so that their combined planform area remains within the 30% wing-area constraint, using simple balsa structures that are easy to manufacture yet sufficiently strong for repeated hand launches.



Figure 1: Image of Glider

2.1 Material Selection

The material selection was crucial for meeting the glider's main objective of travelling as far as possible with the least weight possible. To meet these objectives the materials selected must be light weight, they must have the ability to be processed into an aerodynamic shape, they must have the strength to support the wing, and the impact resistance to withstand test runs. Therefore, before making any decisions regarding the geometrical design of the glider, it was necessary to have a thorough understanding of the materials available for construction. First, each material was researched to determine its physical properties, which provided insight on their potential applications as well as how well they aligned with the glider's objectives. Each materials manufacturability was also researched to ensure that any design that was considered was feasible given the materials processing restrictions. The information gathered is summarized in table 1.

Table 1: Material properties and manufacturability

Materials Provided	Fulfillment of design requirements	Process design Considerations
Rigid Foam	The least dense material available with a high strength to weight ratio. Highly impact resistant, would be useful to prevent failure upon landing. It has a rough surface which could increase the gliders drag	Can be formed by cutting, sanding, taping, gluing. It can even be heat treated to increase its moldability. It doesn't require mold. Cannot be formed with high complexity.
MDF ~ Medium-density fibreboard. Made by breaking down hard or softwood into wood fiber. fiber reinforced secondary, binded with polymer adhesive.	MDF was denser than all materials except ABS. Material properties are Anisotropic thus its strength and density may vary across its surface and cross section. MDF has a hard flat smooth surface, flexible. Not water resistant, fine for indoor applications. Strong surface. Weaker less durable	Not suitable for drilling as it is susceptible to splitting. One advantage for MDF is edge machinability although any manufacturing processes may lack accuracy and detail.

	than plywood	
High grade Plywood	It is one of the heaviest materials available. The direction of the grain changes in each ply of veneer thus the plywood has nearly equal strength properties along its length and width . Good torsional stiffness.	It is formable along the grain and could be bent into curves. It's susceptible to splintering when drilled and milling or cutting may leave rugged surfaces, risk of top layer tear out when milling. It can be laser cut, and joined using tape, glue, or epoxy.
Grade A Balsa wood: 1/16 and 1/8 in sheets.	Has isotropic density. Compressive strength: 7MPa, E = 3Gpa, TS = 14MPa. Very strong bending strength to weight ratio, able to withstand high lift force. Extremely thin therefore light but very highly warped.	Easily cut and shaped, may be difficult to smooth as the end grain tends to fuzz. Will need to be trued to achieve a straight edge. May bow (bend/warp) when cut. Cannot hold nails, can be joined by glue or tape. Can be lasercut
Coreplast ~ Corrugated Plastic: Fluted Polypropylene Sheet	Somewhat light weight heavier than display board, shockproof, waterproof. Not rigid, prone to warping, durable, and can withstand crashes better than balsa wood.	Easily cut with a utility knife, Coreplast cannot be joined by glue but can be joined via laminate. Will fold easily when scored.
Display Board ~ Cardboard and Foam	It has a high rigidity, light weight but heavier than balsa. Lower strength to weight ratio compared to balsa. Good impact resistance.	The display board is malleable if the inner layer of the card board is removed, but it would lose strength. It can be lasercut, but it will melt the material, thus a tolerance will have to be taken into account. It does not join well with glue. It can also be formed by scoring and

		bending. It can be joined with tape.
CardBoard ~ Corrugated B-flute: 3.2mm-1/8" in thickness	Similar to the display board, but less rigid and more likely to warp. Good impact resistance and may make a good reinforcing agent. The density was not provided but it was calculated to be the lightest material available ranging from 0.02-0.009 g/cm^2	Cannot be reliably bent, but can be laser cut. Can be joined by glue, epoxy or tape.
Blue SF ~ Polystyrene	Hard, smooth and light, but brittle. It may be useful to smooth the rough leading edges of the wing and tail planes. Blue in colour, could also be used for aesthetic purposes	Can be extruded and vacuum formed. Can be molded when heated, its rigidity is maintained when cooled. Can cast into molds with fine detail, useful for higher complexity designs. Temperature can be controlled using photocrosslinking i.e. UV light.
ABS~Acrylonitrile butadiene styrene	Densest material making it impractical for a lightweight design. It has high impact resistance and fatigue resistance. The material is hard and rigid, stronger than pure polystyrene, tough, ductile, and is able to maintain an excellent stability under load. Known to warp due to shrinkage during its cooling process	Can be 3D printed allowing for highly complex designs. Can be used for small parts with a minimum height and width of 0.02" wX0.01" h. Desirable properties obtained from molding at low temperatures, but this may not be an option. It is easily machined by turning, drilling, milling, and sawing. ABS parts can be welded together by

		heating the joint surfaces until they begin to melt; Reinforcement can be applied to such a joint by melting a thin ABS rod. ("ABS Plastic Sheet, Rod, Tube and Accessories". Interstate Plastics. Retrieved Nov 17, 2025.
Support beams		
Basswood stick 0.5"x0.25"x36"	26.1g so considerably heavier	Can be cut to desired length
30cm Bamboo skewer 3.1" vs 3.7" diameter	1.77-2.65g each, lightest option provided	Can be cut to desired length
Carbon tube 50cmx7.90mmDx6.8mmID ~extruded CF	8.5g, sufficiently light, strong and rigid.	Length cannot be adjusted but the tube is hollow, thus the tailplane could be inserted.

2.1 Wing

The provided composite wing was the main constraint on wing design, and thus our considerations were mostly on the integration, mounting, and stability-related decisions rather than changing the airfoil or planform. With those things in mind, the baseline parameters used to make design decisions were 29.5 in span, 4 in chord, and 108 in² surface area. Because wing performance is highly dependent upon consistent angle of attack and alignment at launch, the most important design choice was to use a mounting approach that is secure but still allows adjustment during trimming. The attachment style using a skewer and elastic band was chosen because it holds well but still makes it easy to shift the position of the wing slightly when doing test runs.

Several wing geometry features were considered as possible performance enhancements. Dihedral was considered because even a few degrees can noticeably improve lateral stability. Winglets were also considered, as blended winglets can reduce vortices and induced drag, and they are able to improve sink performance at low speeds; however, they also add build complexity and require accurate alignment in order not to create extra drag or imbalance.

Wing sweep was also avoided as a design feature, as sweepback is mainly useful for high-speed aircraft and tends to worsen stall and low-speed performance - definitely not something desirable in a hand-launched glider.

In the end, the wing was subjected to a secure finishing procedure and its weight was determined prior to and post-trimming in order to verify that its mass is in the expected range. One of the paramount CG trim goals that was constantly applied during the tests was to confine the CG within the region from 25% to 33% of the wing chord, which is more or less equal to 1.0 to 1.33 inches from the leading edge for a 4 inch chord.



Figure 2: Image of the Composite Wing

Considerations

The wing was positioned dihedrally meaning at a slight angle of attack increases stability and helps the glider to fly straight. Dihedrals improve the aircraft's lateral stability, a few degrees can produce significant improvement. The wing was attached on top of the fuselage using skewers through the center of the fuselage and elastic bands securing it in place preventing twisting and sliding. The wing was positioned accordingly to balance the center of gravity. The CG should be located between 0.25 and 0.33% of the chord from its leading edge. Therefore 1-1.33 inches from the wings leading edge [12]. This was taken into consideration during manufacturing

2.2 Fuselage

To facilitate the mounting loads and ease of manufacture within a short time, the fuselage had to be lightweight, strong enough, and the fuselage material concept options included solid balsa, laminated balsa, a 3D printed ABS nose, and various combinations of display board-based structures

with reinforcement. A smooth, narrow fuselage profile will minimize drag and allow adequate space to adjust ballast to permit accurate trimming.

The final approach was to use a display board to cover the rigid foam block approach which is a proven design method for the construction of a fuselage. The use of display board was selected because of its lightweight nature and ease to mold into a shape, also contributing to a smoother outer surface for reduced drag. Rigid foam shaped the nose geometry and wing saddle of the internal structure to prevent local deformation where the wing mounts to the fuselage.

To support trimming, batteries were located behind the nose as adjustable mass ballast, which allowed the CG to be shifted forward or backward during flight tests without adding superfluous structure. This makes iterative testing easier and improves repeatability. A carbon fiber tube concept was also employed as a strong, light backbone to support the fuselage and tail while meeting length constraints with minimal weight increase.

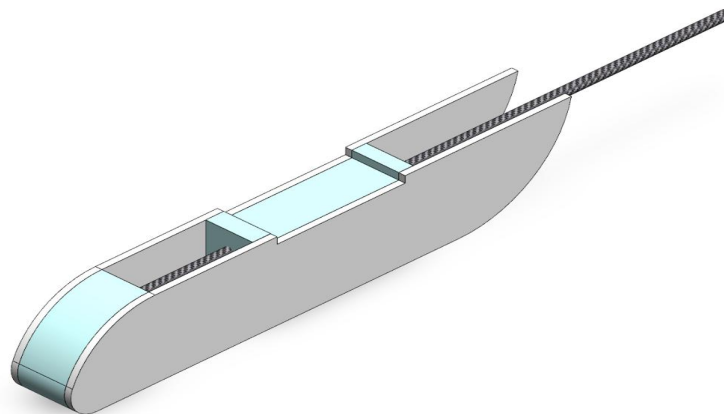


Figure 3: CAD model of the glider fuselage showing the internal foam supports and carbon fiber tail boom.

Fuselage specifications

- Length: 4-6 times the chord of the wing, therefore between 16 and 24 inches OR 75% of the wingspan so 562mm (22 inches) [11]. Carbon fiber rod ~ 50cm or 19.7in
- Slenderness ratio should be 1:6 or 1:5 of the respective diameter and length of the fuselage (d_{max}=4.8in, d_{min} = 2.7in or 68.5mm [10])

2.3 Empennage

The empennage provides pitch and yaw stability, is light weight and meets the tail sizing limits of the project. With a wing surface area of 108 in², the total tail area target must be less than 30% of the wing area; therefore, the tail's planform area cannot exceed 32.4 in².

Tail sizing and geometry have been determined by utilizing standard sizing rules found in the reference report along with applying traditional sizing methods for tail design, such as targeting tail volume coefficients in the range of $V_{ht} = 0.5$ and $V_{vt} = 0.09$, and making the empennage suitable for low-speed flight. Key stability-related recommendations included the use of a symmetrical tail airfoil, a rounded leading edge with a sharp trailing edge, and setting the fixed horizontal tail to a small negative incidence (about 2 to 3 degrees down) to create stabilizing negative lift when required.

For the final configuration, the stabilizers were constructed from 1/8 inch balsa, selected because it met the thickness expectation and had a good stiffness-to-weight ratio. The selection of a conventional tail over a T-tail has made a more stable choice with regard to stall characteristics at low speeds and also has decreased the potential for flow blanketing due to the tail fin location. Tail structure was supported using a carbon fiber tube boom, selected because it was stiff, light, and allowed a longer tail moment arm without significant increase in weight.

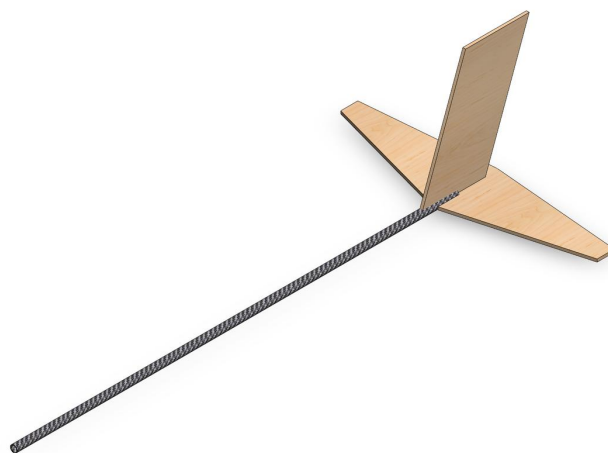


Figure 4: CAD model of the empennage assembly showing the horizontal stabilizer, vertical stabilizer, and carbon fiber tail boom.

Material Options

The materials considered for the tail plane were plywood, 1/16 balsa as well as 1/8 balsa wood. The 1/8 balsa was chosen as it provides structure and strength needed to resist impact while remaining

lightweight. The 1/16 balsa is too light and will not stand stress tests. The plywood has the strength but adds to the weight.

Table 2: Tailplane selection considerations

Tail Planes	Advantages	Disadvantages
T-tail	Offer greater stability and reduced drag because horizontal stabilizers are located away from the turbulence wakes of the wings (less downwash). Increased pitch control, good glide ratio, more efficient for a low speed aircraft. Less buffeting	Likely to stall at high angles of attack, may increase the risk of flutter. Difficulty recovering from spin. Must be stronger and heavier than a conventional tail. Smaller than conventional tail.
Cruciform	Compromise between T-tail and Conventional, weight lies between the two.	Small surface area
Fuselage mounted ~Conventional tail	Most light weight, appropriate stability. Stabilizer trim can be added while maintaining low complexity.	May experience blanketing of the vertical tail leading to loss of yaw stability, this is more likely to occur at a high angle of attack. (Can be prevented by placing the horizontal tail further behind the vertical tail) Larger vertical tail may also help. Larger horizontal surface area advantage
V-Tail	Requires 70.7% of the tail area of the conventional tail in theory	Mechanics required for rudder and elevator inputs. Rudder deflection of v-tail may cause a roll moment

Inverted V-tail		Rudder deflection avoided with inverted V-tail
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Table 3: Tail plane possible features

Fin Types	Advantages	Disadvantages
Twin Tail	Covered less by the fuselage in cases of high angles of attack and therefore advantageous in this condition	Structural complexity and extra weight
Triple fin	Provides yaw stability at higher angle of attack	Heavier due to extra components
Ventral fin	Not blanketed with high angles of attack, prevents lateral instabilities in high speed flight	Ground strikes could damage the fin
Dorsal fin	Improves efficiency of tailplane where high angles of yaw exist, stall moved to higher angles of yaw	May need a stronger fuselage

Tailplane Specifications

- Aspect ratio of horizontal tail plane should be half that of the wing [1]
- Sweep of the horizontal tailplane should be 5° larger than the sweep of the wing [1]

- Sweep angle of the vertical tail plane should be less than 20° for low airspeeds without compressibility effects [1]
- Should have a relative thickness approximately 10% less than that of the wing [1]
- Vertical tailplane(s) should have symmetrical airfoil, rounded blunt leading edge, and a sharp trailing edge [9]
- Horizontal tailplanes should be symmetrical or almost symmetrical, asymmetrical airfoils are installed upside down because horizontal tailplane creates negative lift [9]
- Since the horizontal tailplane will be fixed it must have an incidence angle of around 2° to 3° downwards to create negative lift ~Could consider a moveable horizontal tail wing that is adjusted with test runs [9]
- For sailplanes conventional tail lever arms (distance between aerodynamic center of the main wing and the aerodynamic center of the horizontal tailplane) of horizontal and vertical tails should be 65% of the fuselage length [1]
- * Relative thickness refers to the ratio of the airfoils maximum thickness to its chord length

Tail volume coefficients: $V_{ht} = 0.5$ $V_{vt} = 0.09$

2.4 Final Design

Multiple design options were considered before the final design was chosen. The tailplane was decided to be 1/8" balsa wood, as the thickness met the required tailplane specifications, and it provided the best strength to weight ratio of all other materials. After some debate between using a conventional and T-tail plane, it was decided to use a conventional tail to avoid the high risk of dead stalls at low speed. The vertical tail plane was also placed in front of the horizontal tail planes to avoid blanketing. A circular fuselage design made of display board and cardboard supports was considered, as it was highly aerodynamic and extremely lightweight, but it was scrapped due to its complexity making it difficult to manufacture with limited time. It was therefore decided to create flat display board skin, attached to a rounded rigid foam nosecone and wing support. The skin allowed the use of less foam board, created space for a removable ballast, and stronger wing support with easily adjustable wing positionings. The Foam nosecone and wing support naturally acts as a bumper with high impact resistance, making the glider sturdy upon testing. Finally it was decided to support both the fuselage and tailplane on the carbon fiber tube, as the foam was easily drilled to fit around the tube, and it allowed for a glider length that met specifications without adding significant weight.

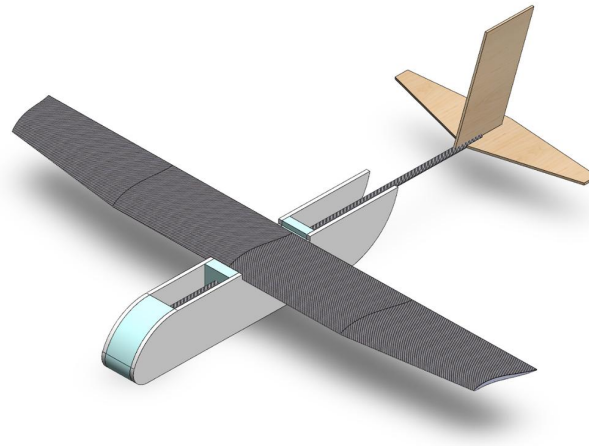


Figure 5: Overall CAD model of the hand launched glider. The composite wing is mounted to a lightweight fuselage, with a carbon fiber boom supporting the tail assembly.

3.0 Wing Design Analysis

3.1 Wing Parameters

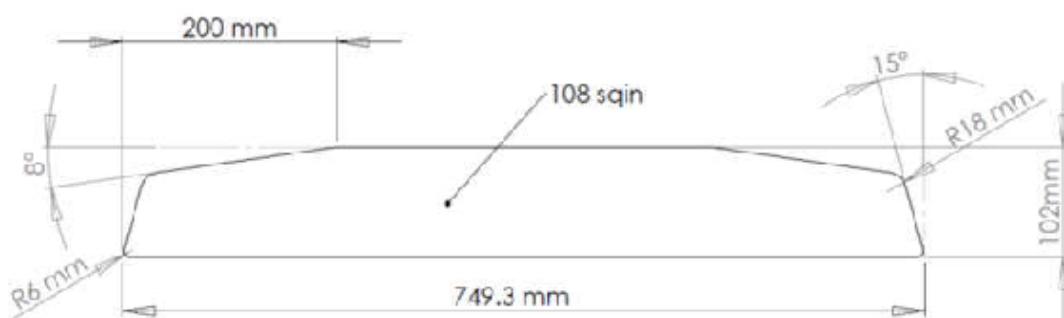


Figure 5: Wing Shape

Table 4: Wing Parameters

Span	Chord	S.A.
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29.5 in	4 in	108 in ²
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Table 5: Mass of Composite wing and glider

Mass Carbon fibre and Fibre Glass	Total Weight of Wing	Weight of Epoxy	Weight Glider	Weight Payload
21 g	54.9g	33.9g	175g	200g

3.2 Airfoil Selection

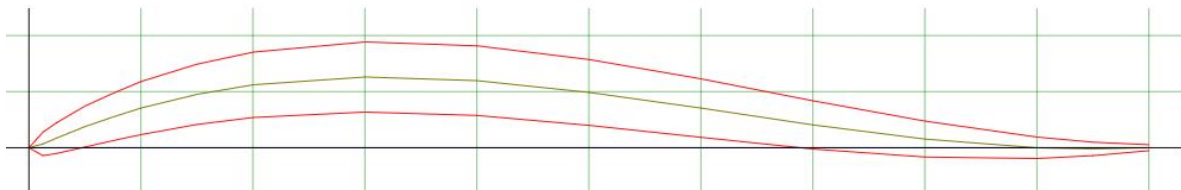


Figure 6: NACA M22 Airfoil [6]

The NACA M22 airfoil was chosen for the glider. This airfoil is a thin airfoil with a max thickness of 6.3% and 30% of the chord. The M series airfoils are modified and designed for improved stall behaviour. Figure x shows the lift for different angles of attack for increasing Reynolds numbers. As seen the C_l vs AoA line remains relatively similar for all Reynolds numbers. This is because the M series airfoils are optimized for high lift and low Reynolds numbers. The M series airfoils are mainly used in small aircraft applications like gliders. The airfoil has gentle stall as seen around the angle of 10-12 the lift slowly decreases and it reaches a higher $C_{l_{max}}$.

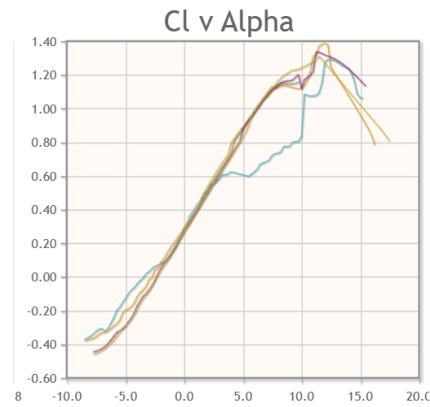


Figure 7: CL vs AoA [6]

3.2 Material Selection

To Select the best material for the wing it was modelled as a cantilever beam using half of the wing to model the stress due to the lift required for the glider.

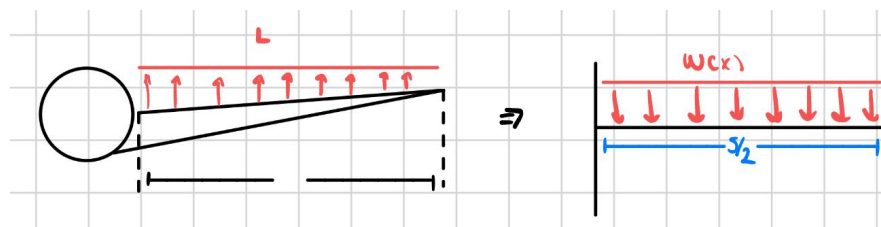


Figure 8: wing modelled as cantilever beam

The maximum bending stress due to lift on the wing was found to be 1.221 Mpa. The density of the composite wing was calculated as 0.00156 Mg/m^3 . These were used to find the specific strength which is 782.69 and then select a material using the selection chart below

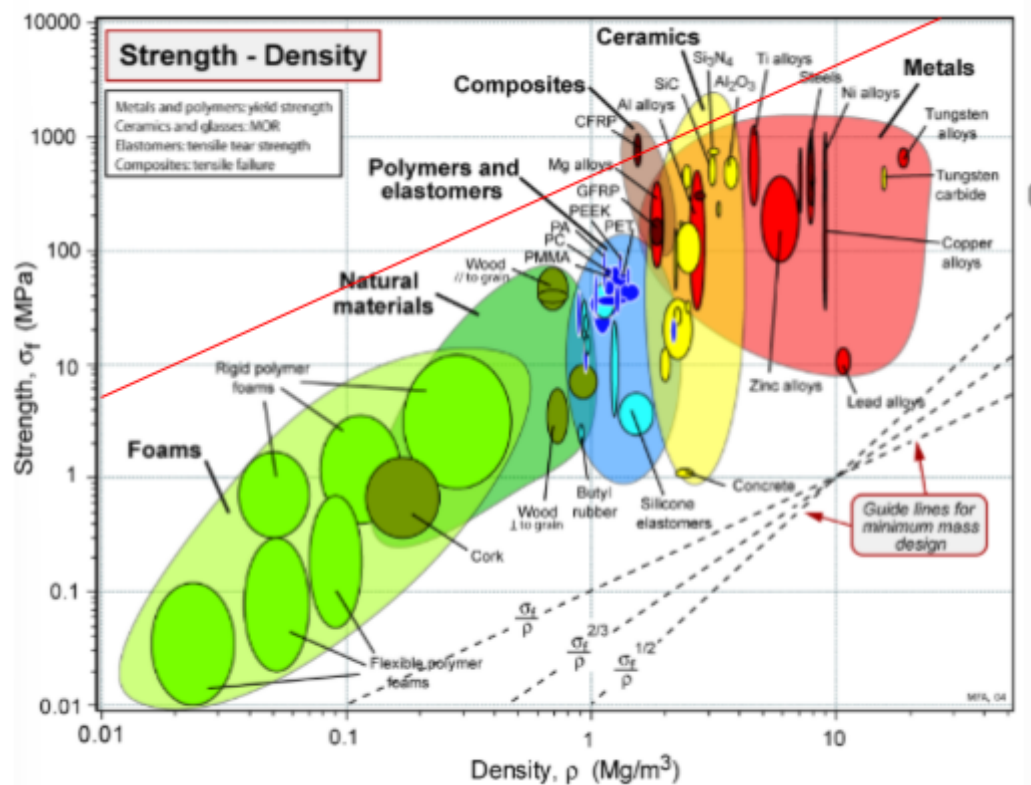


Figure 9: Material Selection Chart

The red line in Figure x above is the line of the calculated specific gravity using the half wing approximation. It can be seen that the ideal material is CFRP composites which is what has been used for the wing in manufacturing. These composites have a high tensile strength and stiffness while remaining lightweight. They are typically a carbon Fiber and Epoxy mix for our wing Fiber glass has also been used making it a hybrid CFRP. This material meets the goals of the glider ensuring sufficient strength for the lift required while remaining light weight.

The Weight of the composite wing was measured to be 54.9g from this the volume fractions of the components were found and the material properties of the composite were calculated. The modulus of elasticity of the composite was found to be 12.997 GPa which is not as stiff as metals but has a similar stiffness to wood. It allows for some bending while remaining stiff enough to resist deflection. The Tensile strength of the composite was found to be 0.509 GPa which is a sufficient strength for the wing of the glider.

The Payload to weight Ratio was found to be 0.533. Ideally the ratio should be as close to 1 as possible as the higher the ratio the better your aircraft is at carrying weight. A higher ratio shows higher lift efficiency and overall will result in a better flight score. Overall our ratio is acceptable but has a lot of room for improvement. Overall The wing design has been chosen to best meet lightweight and strength requirements.

4.0 Manufacturing Design

The manufacturing design for the glider focuses on producing a lightweight, structurally stable aircraft structure using the materials provided in the laboratory. The main components; the fuselage, tail, nose, and the wing, were developed to ensure efficient manufacturing, light weight, and structural durability. The processes described below reflect practical fabrication steps that align with the capabilities of the manufacturing lab.

4.1 Mold-Based Manufacturing

The wing was fabricated using a two piece mold that was created from a reference foam wing. The foam plug captured the full profile of the airfoil, allowing the molded composite wing to replicate the geometry precisely. Both the upper and lower mold halves were cleaned, polished, and coated with several layers of release wax to prevent adhesion during curing.

A 50 g shot of 2.0 lb/ft³ polyurethane foam was poured into the closed mold to form the internal wing core. After expansion and curing, the mold was separated and the foam core was removed. Excess flashing was trimmed off, and the core was sanded to the parting line for a clean finish inside the composite skins. Additionally weight was reduced by laser cutting holes into the foam, creating internal cavities that also served as spars and pockets for fiberglass ribs. This method produced a lightweight but stiff internal structure tailored for the thin wing profile.

4.2 Composite Wing Layup

A wet layup method was used to manufacture the composite skins. The upper surface of the wing was constructed from 2.9 oz carbon fiber cloth to maximize stiffness, and the lower surface used 2.0 oz fiberglass cloth to reduce weight. Both cloth layers were cut to shape and weighed prior to resin application.

Epoxy resin MVS-410 and hardener MVS-464 were mixed using a measured ratio such that the total epoxy mass was approximately twice the mass of the fiber materials that achieves a workable viscosity and pot life. For this layup, 33.4g of resin and 5.8g of hardener were used, A thin bead of epoxy was applied to each cloth layer length wise and distributed evenly with squeegees to ensure

complete wetting while avoiding excess resin. Fiberglass strips (1.6 oz) were placed along the leading and trailing edges to increase the bond strength between the two skins.

The foam core was centered in the upper mold and pressed gently into place. The edge strips were folded over the core to create a continuous shape. Fiberglass rib pieces were wetted and applied into the internal recesses to stiffen the wing across its span. Once all reinforcements were installed, the lower mold was aligned and bolted onto the upper mold. The assembly was left to cure completely before demolding.

The cured wing was trimmed using a diamond blade saw, and the edges were sanded for a smoother finish and to achieve the final aerodynamic profile. The wing was weighed before and after trimming to ensure the weight remained within acceptable limits and consistent with the design calculations.

4.3 Fuselage Design and Manufacturing

The fuselage was constructed using a display board as an outer skin that is wrapped around internal rigid foam blocks. The display board was chosen because it is extremely lightweight, easy to shape, and provides a smoother surface, which benefits the aerodynamic performance. The outer fuselage panels were cut, scored, and curved into shape before being bonded in place with epoxy.

Rigid foam sections were carved to form both the wing saddle and the rounded nose geometry. The foam also acted as the primary internal structure, preventing local deformation and providing a solid surface for the wing to rest against during mounting.

A few batteries were incorporated directly behind the nose block as adjustable mass to fine-tune the center of gravity. Their position could be shifted forward or backward within the slot, allowing the CG to be adjusted accurately during test flights without adding unnecessary structural weight. The foam interior ensured the batteries remained secure during launch and landing.

This method produced a fuselage that was lightweight, a stable structure and was easy to modify during iterative testing.

4.4 Tail Design and Manufacturing

The tail assembly was fabricated from 1/8-inch balsa wood, selected for its stiffness-to-weight ratio and its ease of machining. Templates were drawn for both the horizontal stabilizer and the vertical fin, and the components were cut, with all edges and surfaces sanded to achieve smooth, symmetric aerodynamic profiles.

A flat-plate airfoil was used for the horizontal stabilizer, providing a small amount of lift. Both tail components were bonded to the carbon-fiber tail boom using epoxy, offering added strength without a significant mass increase.

The result was a lightweight, stiff tail assembly that ensured stable flight characteristics while maintaining ease of manufacture.

4.5 Assembly Process

The final assembly involved combining all the components; the wing, fuselage, and tail into a complete glider. The carbon-fiber tail boom was inserted through the rear fuselage foam block and aligned carefully before being permanently bonded. The tail components were attached and checked to ensure they were symmetrical relative to the fuselage centerline and the wing plane.

The composite wing was mounted using bamboo skewers inserted through the fuselage sides and tightened with rubber bands. This allowed easier adjustment of the wing's position and angle during testing and provided a release mechanism to minimize damage in the case of a hard landing.

During assembly, batteries were installed in the slots behind the nose to achieve the correct center of gravity location. Their position was fine-tuned until the glider balanced between 25 percent and 33 percent of the wing chord.

4.6 Design Inspection

We inspected the glider to make sure it met the size limits, was strong enough, and was properly balanced before testing. We did this in three steps: after making the parts, after assembling everything, and right before the flight tests.

Wing inspection (composite):

- Verified that the wing was trimmed and sanded to a smooth finish with minimal flaws such as cracks, delamination, or irregular edges. The wing weight was also checked to confirm it stayed within the expected range before and after trimming.
- Verified that the wing could be mounted securely using the bamboo skewer and rubber band method, without excessive movement at the mounting points.

Fuselage inspection (display board and rigid foam):

- Verified that the fuselage skin was properly bonded to the internal foam blocks, with closed seams and a smooth outer surface to minimize drag.
- Verified that the wing saddle area was stiff enough and did not crush locally when pressed by hand, with the foam providing adequate support.

Tail and boom inspection (balsa tail on carbon tube):

- Verified that both tail surfaces were sanded smooth and kept symmetrical to avoid unwanted yaw or roll coupling.
- Verified that the balsa tail components were bonded to the carbon fiber boom using epoxy, and that the boom was straight, aligned, and bonded through the rear fuselage block.

Mass balance inspection:

- The center of gravity was set by adjusting the internal battery ballast until the glider balanced between 25 percent and 33 percent of the wing chord, which is about 1.0 to 1.33 inches from the leading edge for a 4 inch chord.
- Final mass was recorded to confirm it met the build mass requirement, with a total glider weight of 175 g.

Final alignment checks prior to flight:

- Verified that the tailplane was level relative to the wing plane, and that the fin was perpendicular to the tailplane.
- Verified left right symmetry by checking that the tail was centered on the fuselage centerline with no visible twist. Also confirmed that all glued joints were fully cured and solid, and that there were no sharp edges.

4.7 Comparison to Alternative Methods

The final configuration of this glider was reached after considering several alternative methods of building the glider. For the fuselage, the choices were solid balsa wood, laminated balsa wood, an ABS 3D printed nose, and different types of structures using display boards with internal reinforcements. Solid or laminated balsa wood is very strong and simple to construct, but if you need additional thickness for it to remain stiff enough, those fuselages could end up quite heavy. The ABS 3D printed nose will yield a rounded profile and improve the crash resistance of the fuselage, but it does add a manufacturing complexity and may add to the weight or create warpage if not properly created. The method chosen allows for a lightweight design, is quick to manufacture, and can be easily modified during the trimming and testing phases.

Various options for the tail were evaluated prior to selecting an eventual configuration. The tail configurations included: T-tail, cruciform tail, conventional tail, and V-tail. While the T-tail may increase pitch control; because there needs to be a stronger and heavier vertical tail, they can be very difficult to recover from high angles of attack. While the theory is that the V-tail can reduce the necessary tail area, its use requires mixing the control inputs and can result in roll coupling when deflecting the rudder. The conventional tail configuration was preferred because it is lightweight, provides a stable platform, and produces a simple overall design. Tailplane Materials Selection; 1/8" Balsa because of strength to weight ratio and ability to perform as required by tailplane.

Wing Optional Features; Winglets; see comments above. Blended winglets reduce vortex generation and drag as well as increase the low speed sink performance of a wing; however, blended wings increase building complexity and increase sensitivity to alignments of all parts. Wing Integration; Skewer and Elastic Band Method; Provides secure wing attachment, adjustable during test flight, and reduces risk of destruction during crash landings.

5.0 Discussions

After accomplishing flight distances of over fifteen metres and showing excellent consistency throughout multiple launches, it can be inferred that this glider was trimmed well based upon the mannerly, repeatable, and predictable behaviour of the aircraft during the glide phase following release. The smooth yet predictable glide demonstrated that the glider was properly aligned through launch and glide path, resulting in a stable condition following release from launch. The glider

exhibited no porpoising (pitching up and down) and did not roll off into a large half-loop after release. The fact that the glider behaved consistently and predictably throughout multiple launches indicates that the design is not dependent on the factor of a perfect throw but will always return to the same natural steady glide from any position on the glide path.

One of the most significant factors affecting performance was the trim balance and mass balance, especially regarding the centre of gravity. The batteries that were incorporated into the design as ballast were used to place the centre of gravity within an acceptable range of the wings. If the centre of gravity is positioned too far rearward, pitch sensitivity increases. The glider's nose will tend to rise quickly, causing a drop in air speed and causing the wing to operate at a greater angle of attack (greater induced drag), increasing the likelihood the glider will stall. A stall will cause loss of altitude, resulting in a decrease in range. Using battery ballast to adjust the centre of gravity on this glider led to increased balances (aerodynamic moments) between the tail and wings and thus maintained pitch stability for the majority of the flight without oscillating. Flight quality and uniformity from launch to landing were also enhanced by all components being correctly aligned and by utilising a solidly mounted wing system that prevents any movement of the wing during launch.

Even minor variations in incidence can cause an immediate change in pitch behaviour, so establishing secure wing mounts is vital to maintaining the proper incidence while gliding. Furthermore, having a rigid wing saddle (a wing's mounting surface) keeps the wing in a consistent angle throughout the entire glide phase, making the flights more reproducible than those with a flexible wing saddle. The fuselage of the glider should have a smooth finish (without seams, rough edges or unlevel transitions) since any parasitic drag (caused by an object's shape) will reduce the glider's ability to convert its altitude into horizontal distance (speed) as efficiently as it would otherwise. Tail symmetry and alignment have equal importance. Any tilt in the fin or a non-parallel tailplane will create a continuous slow turn that increases drag, and decreases flight range (distance) of the glider.

The main limitations of this system are the hand-launched aspect of it, and how sensitive the system is to very small geometric inaccuracies. The angles and speeds at which the glider is launched by hand differ, which leads to changes in launch range of many meters depending on the hand launch angle and hand launch speed. Added to this issue are added weight caused by using a large amount of epoxy at the glued joints. When there is enough strength at the joints of a glider, the glue is there just to hold everything together as opposed to providing additional performance. Future improvements should be focused on ways to simply align the joints during the curing process with an

alignment fixture, and reduce the amount of glue required to make the joint stronger, as well as adding trim tabs or finer ballast adjusters to make it easier to adjust the trim of the glider after it is completed.

6.0 Conclusions

This project has led to the creation of a glider which is not only light but also stable and within the limitations of our designs and, at the same time, very good in terms of flight performance. The final configuration includes a composite wing and a fuselage and tail structure that is both simple and easy to manufacture, thus allowing for low weight, high stiffness, and accurate alignment. A structured design review was conducted to verify correct wing attachment, fuselage structural integrity, tail alignment, and proper mass balance prior to performing any flight testing.

Flight test results confirmed all of the design choices made for this project. After trimming the glider, it successfully achieved maximum distance flown of 15m+ with identical and repeatable glide characteristics. This was primarily possible due to the correct placement of the centre of mass (using battery ballast), maintaining a rigid wing saddle to preserve wing incidence angle, and ensuring that the tail was both symmetrical and square relative to the fuselage in order to prevent unnecessary drag during yaw or roll coupling.

Overall, this project achieved its goal of designing an efficient and reliable glider within the materials and methods of production that were available. Future enhancements will focus primarily on the repetition and the range increase, that is, by decreasing the excessive epoxy amount that is used during the assembling process, the employment of simple jigs to help the future assemblies and the introduction of the possibility of making very small trimming adjustments by using tiny trim tabs or positioning of ballast very precisely.

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Appendix A- Wing Design Analysis Calculations

Wing Parameters

NACA M22

Span=s=29.5in

c=4in

S.A=108in²

Weight of carbon fiber and fiber glass : 21g =0.0462971lb

Weight of wing = 54.9g =0.1210338 lb

Weight epoxy= 54.9-21=33.9 g=0.07474 lb

Carbon/Epoxy Ratio Ideally 60/40

The following approximation was used, using NACA M22 characteristics from figure 1 in appendix 1:

For typical airfoils

$$k_l = 0.036$$

$$I = K_l c t (t^2 + h^2) = 0.036 * 4in * 0.25 * (0.25^2 + 0.285^2) = 5.174e - 3in^4 = 2.1519E-9m^4$$

$$z_{NA} = \frac{z_u - z_l}{2} = \frac{10.25 + 0.02}{2} = 5.135$$

$$t = \max\{z_u - z_l\} = 6.25\% \text{ of chord}$$

$$t = 0.0625 * 4 = 0.25in = 0.00635m$$

$$h = \max\{(z_u + z_l)/2\} = 7.125\% \text{ of Chord}$$

$$h = 0.07125 * 4 = 0.285in$$

$$A_{\text{Airfoil}} = K_A * c^2 * (t/c) = 0.6 * 4^2 * (0.25/4) = 0.6in^2$$

$$V_{\text{wing}} = A_{\text{airfoil}} * s/c = 0.6 * 29.5/4 = 4.425in^2$$

Table Ax: Material properties

	Fiberglass Cloth	Epoxy Resin
SG	2.5	1.13
Tensile Strength	TS 2.0 Gpa	68.6 MPa
Modulus of Elasticity	E=72 Gpa	3.31 GPa
Density	2.5g/cm ³	1.13g/cm ³

Assuming all Carbon + fiber glass using Fiber glass properties

$$\rho = SG * 1g/cm^3$$

Finding Volume Fractions:

$$V_f = \frac{m_f}{\rho_f} = \frac{21}{2.5} = 8.4 \text{ cm}^3 = 8.4E-6 \text{ m}^3$$

$$V_m = \frac{m_m}{\rho_m} = \frac{33.9}{1.13} = 30 \text{ cm}^3 = 3E-5 \text{ m}^3$$

$$V_t = V_f + V_m = 3.84E-5 \text{ m}^3$$

$$f_f = \frac{V_f}{V_t} = \frac{8.4E-6}{3.84E-5} = 0.21875$$

$$f_m = 1 - f_f = 1 - 0.21875 = 0.78125$$

Density of composite:

$$\rho_c = \rho_m f_m + \rho_f f_f = 0.21875 * 2.5 + 0.78125 * 1.13 = 1.5625 \text{ g/cm}^3 = 0.0015625 \text{ Mg/m}^3$$

Bending Stress using half wing approximation:

$$L = s/2 = 29.5/2 = 0.37465 \text{ m}$$

$$m_{\text{wing}} = 54.9 \text{ g} = 0.0549 \text{ kg}$$

$$M_{\text{empty}} = 175 \text{ g} = 0.175 \text{ kg}$$

Design load factor = $n=3$: ensuring sufficient lift

$$g = 9.81 \text{ m/s}^2$$

$$W_{\text{Glider}} = 0.175 * 9.81 = 1.4715 \text{ N}$$

$$L_{\text{Design}} = n * W = 3 * 1.4715 = 4.4145$$

$$F_{\text{Half}} = L/2 = 2.20725 \text{ N}$$

$$w(x) = F_{\text{Half}}/L = 5.895 \text{ N/m}$$

$$M_{\text{Max}} = \frac{w(x) * L^2}{2} = \frac{5.895 * (0.37465)^2}{2} = 0.4137 \text{ N} \cdot \text{m}$$

M_{max} @ root

$$\sigma_{\text{max}} = \frac{M_{\text{max}} * z}{I} = \frac{0.4137 * 0.00635}{2.1519E-9 \text{ m}^4} = 1220779.3 \text{ Pa} = 1.221 \text{ Mpa}$$

$$\text{Specific Strength} = \frac{\sigma}{\rho} = \frac{1.221 \text{ Mpa}}{0.00156 \text{ Mg/m}^3} = 782.69.231$$

Composite Material Properties:

$$E_c = \frac{E_f E_m}{f_m E_m + f_f E_f} = 12.997 GPa$$

$$\alpha = \frac{E_m f_m}{E_f f_f} = 0.1642$$

$$TS_c = (\alpha + 1) TS_f = 1.1642 * 0.21875 * 2.0 = 0.509 GPa$$

Payload Ratio:

$$Payload\ ratio = \frac{W_{payload}}{W_{wing + fuselage + payload}} = \frac{200g}{175g + 200g} = 0.533 = 53.3\%$$

Appendix B - Design Calculations

Conventional Tail Plane Dimensions

Given Requirements:

1. The surface area of the tail plane must be at most 30% of the wings surface are

$$(1) \quad A_{tailplane} = 0.3 A_{wing}$$

The area of the wing was given as 108 in^2 , thus the surface area of the horizontal tailplanes can be calculated as can be calculated as follows

$$A_{tailplane} = 0.3(108) = 32.4 \text{ in}^2$$

2. The Aspect ratio of the tailplanes should be half that of the wing

$$(2) \quad AR_{tail\ plane} = \frac{1}{2} AR_{wing}$$

Where the aspect ratio is defined by the wings span squared divided by its surface area

$$(3) \quad AR = \frac{S^2}{A}$$

The surface area of the wing and horizontal stabilisers have been determined, and the span of the wing was given as $S = 29.5in$, the desired wingspan of the horizontal stabilizers are determined as follows.

$$S_{tailplane} = \sqrt{(A_{tailplane})\left(\frac{1}{2} \times \frac{S_{wing}^2}{A_{wing}}\right)} = \sqrt{(32.4)\left(0.5 \times \frac{29.5^2}{108}\right)} \approx 11.425in$$

Note that the semispan of the horizontal tailplanes will be equal to the span of the vertical stabilizer $\approx 5.7125in$

3. Knowing the desired span and surface area of the tailplanes, the standard mean chord (SMC) is calculated using the relation

$$(4) \quad SMC = \frac{A}{S}$$

Thus, the desired standard mean chord of the tailplane should be $SMC \approx \frac{32.4}{11.425} = 2.836in$

4. The sizing of the relative thickness of the tailplanes was determined to be either 10% less than that of the wing, or 9-12%. Where relative thickness is defined as the ratio between a wing's thickness (t) and its chord (c).

$$(4) \quad t_{relative} = \frac{t}{c}$$

With conflicting findings, and sizing limited to the thickness of the material selected, it was decided to create a range of values the relative thickness should lie within.

First the relative thickness of the wing was determined to be 6.3% [7], thus the relative thickness of the tailplane (being 10% less than that of the wing) is calculated to be 5.67%, from this finding, the thickness of the tailplane should be

$$t = t_{relative} \times SMC = 0.0567 \times 2.836 = 0.161in$$

Given the relative thickness of 9-12%, an upper limit can be determined as follows

$$t = t_{relative} \times SMC = 0.12 \times 2.836 = 0.34in$$

Therefore the material selected for the tailplanes should have a thickness of 0.161-0.34 in

Fuselage geometry and wing positioning

The length of the fuselage was decided to be 21 inches, as it was in the desired range of 16-24, and it was a feasible length given the materials provided. Based on this, the desired radius of the fuselage (if the shape was to be a full revolution, desirable for its aerodynamic properties) was determined based on a slenderness ratio of 1:5 or 1:6

1. Thus the radius should have ranged from $\frac{21}{5} = 4.2in$ to $\frac{21}{6} = 3.5$ inches.

Because these values are outside of the allowable fuselage dimensions, it was decided to use the maximum allowable height and width of 3 and 2 inches respectively. In hindsight, a shorter length of 16 to 18 inches would have resulted in a lighter glider, with an optimal radius of 2.67-3 inches, within the allowable range for this project.

With the length of the fuselage, the position of the wing lever arm was calculated given that its distance from the horizontal and vertical tails should be 65% of the fuselage length.

2. Distance between lever arms: $21 \times 0.65 = 13.65$ in
3. Distance between wing lever arm and leading edge of the wing: $25-33\%$ (Chord length)=
 $0.25(4) \text{ to } 0.33(4) = 1 \text{ to } 1.32$ in (space will be provided for the wing within this range, to determine which allows for optimal flight)
4. Position of the leading edge of the wing from the tip of the nose cone: $21-1.32-13.65=6.03$ in

AER 507 Glider Manufacturing Project Work Distribution Form

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